

Equations of First Order and First Degree

2.1. Differential equation of the first order and first degree.
A differential equation of the type

$$M + N \frac{dy}{dx} = 0,$$

where M and N are functions of x and y or constants, is called a differential equation of the first order and first degree.

We give below some methods of solving such equations.

2.2. Solution of the differential equation when variables are separable.

If an equation can be written in such a way that dx and all the terms containing x are on one side and dy and all the terms containing y on the other side, then this is an equation in which variables are separable. Such equations can therefore be written as $f_1(x) dx = f_2(y) dy$ and can be solved by integrating directly and adding a constant on either side.

Ex. 1. Solve $\frac{dy}{dx} = \frac{1+y^2}{1+x^2}$.

Solution Separating the variables the equation becomes

$$\frac{dy}{1+y^2} = \frac{dx}{1+x^2}$$

Integrating, we get $\tan^{-1} y = \tan^{-1} x + A$

or $\tan^{-1} y - \tan^{-1} x = A$ i.e., $\tan^{-1} \frac{y-x}{1+xy} = A = \tan^{-1} C$ (say).

$\therefore y-x = C(1+xy)$
which is the solution.

Ex. 2. Solve $\frac{dy}{dx} = e^{x-y} + x^2 e^{-y}$.

[Gorakhpur 59 ; Andhra 60 ; Sagar 54]

Solution. The given equation can be written as
 $e^y dy = (e^x + x^2) dx$.

Integrating, $e^y = e^x + \frac{1}{3}x^3 + C$.

Ex. 3. Solve $\sec^2 x \tan y dx + \sec^2 y \tan x dy = 0$

[Nagpur T.D.C. 61 ; Delhi 51]

Solution. Separating the variables, we get

$$\frac{\sec^2 x}{\tan x} dx + \frac{\sec^2 y}{\tan y} dy = 0.$$

Integrating, $\log \tan x + \log \tan y = A$

or $\tan x \tan y = e^A = C.$

Ex. 4. Solve $(y - px)x = y.$

[Saugar 62]

Solution. Equation is $px^2 = y(x-1)$, i.e., $\frac{dy}{dx} = \frac{y(x-1)}{x^2},$

$$\text{i.e., } \frac{dy}{y} = \frac{x-1}{x^2} dx = \left(\frac{1}{x} - \frac{1}{x^2} \right) dx.$$

Integrating, $\log y = \log x + \frac{1}{x} + \log A$ or $\frac{y}{x} = Ae^{1/x}.$

Ex. 5. Solve $y - x \frac{dy}{dx} = a \left(y^2 + \frac{dy}{dx} \right).$

[Saugar 63]

Solution. The equation can be written as

$$\frac{dx}{x+a} = \frac{dy}{y(1-ay)} = \left(\frac{1}{y} + \frac{a}{1-ay} \right) dy.$$

Integrating, $x+a = C \frac{y}{1-ay}.$

Ex. 6. Solve

$$(i) (3 + 2 \sin x + \cos x) dy = (1 + 2 \sin y + \cos y) dx.$$

$$(ii) (e^x + 1) \cos x dx + e^y \sin y dy = 0.$$

[Poona 64]

2.3. Equations reducible to the form in which variables are separable.

Equations of the form

$$\frac{dy}{dx} = f(ax + by + c)$$

can be reduced to an equation in which variables can be separated. What is required is that we put

$$ax + by + c = v,$$

$$\text{so that } a + b \frac{dy}{dx} = \frac{dv}{dx}, \text{ i.e., } \frac{dy}{dx} = \frac{1}{b} \left(\frac{dv}{dx} - a \right).$$

Then the equation becomes

$$\frac{1}{b} \left(\frac{dv}{dx} - a \right) = f(v) \text{ or } \frac{dv}{dx} = a + bf(v),$$

in which variables are separable.

Ex. 1. Solve $\frac{dy}{dx} = (4x + y + 1)^2.$

[Raj. 61 ; Agra 54 ; Gujrat 65, 58]

Solution. Put $4x + y + 1 = v$, so that $4 + \frac{dy}{dx} = \frac{dv}{dx}.$

The equation then reduces to

$$\frac{dv}{dx} - 4 = v^2 \quad \text{or} \quad \frac{dv}{dx} = v^2 + 4.$$

The variables are now separable and we can write $\frac{dv}{v^2 + 4} = dx$.

$$\text{Integrating } \frac{1}{4} \tan^{-1} \left(\frac{v}{2} \right) = x + C$$

$$\text{or } \frac{1}{4} \tan^{-1} \left(\frac{4x + v + 1}{2} \right) = x + C \text{ is the solution.}$$

Ex. (2) Solve $\frac{dy}{dx} = \sin(x+y) + \cos(x+y)$. [Agra B Sc. 67]

Solution. Put $x+y=v$, $1 + \frac{dy}{dx} = \frac{dv}{dx}$.

$$\therefore \text{ equation is } \frac{dv}{dx} - 1 = \sin v + \cos v \quad \text{or} \quad \frac{dv}{dx} = 1 + \sin v + \cos v$$

$$\text{or } dx = \frac{dv}{1 + \sin v + \cos v} = \frac{dv}{2 \cos^2 \frac{1}{2}v + 2 \sin \frac{1}{2}v \cos \frac{1}{2}v}$$

$$\text{or } \frac{dv}{2 \cos^2 \frac{1}{2}v (1 + \tan \frac{1}{2}v)} = dx \quad \text{or} \quad \frac{\frac{1}{2} \sec^2 \frac{1}{2}v dv}{1 + \tan \frac{1}{2}v} = dx.$$

$$\text{Integrating, } \log(1 + \tan \frac{1}{2}v) = x + C, \text{ where } v = x + y.$$

$$\therefore \log[1 + \tan \frac{1}{2}(x+y)] = x + C \text{ is the required solution.}$$

Ex. 3. Solve $(x-y)^2 \frac{dy}{dx} = a^2$.

[Calcutta Hons. 63; Bihar 61; Vikram 65]

Solution. Put $x-y=v$, so that $1 - \frac{dy}{dx} = \frac{dv}{dx}$.

$$\therefore \text{ equation is } v^2 \left[1 + \frac{dv}{dx} \right] = a^2 \quad \text{or} \quad \frac{dv}{dx} = \frac{v^2 - a^2}{v^2}$$

$$\text{or } dx = \frac{v^2}{v^2 - a^2} dv = \left(1 + \frac{a^2}{v^2 - a^2} \right) dv.$$

$$\text{Integrating, } x + C = v + a^2 \frac{1}{2a} \log \frac{v-a}{v+a}$$

$$\text{or } x + C = (x-y) + \frac{1}{2} a \log \frac{x-y-a}{x-y+a} \text{ is the solution.}$$

Ex. 4. Solve $(x+y)^2 \frac{dy}{dx} = a^2$

[Poona 64; Raj. 63; Delhi Hons. 60; Alld 60]

Solution. Put $x+y=v$, so that $1 + \frac{dy}{dx} = \frac{dv}{dx}$.

$$\therefore v^2 \left(\frac{dv}{dx} - 1 \right) = a^2, \quad \frac{dv}{dx} = 1 + \frac{a^2}{v^2} = \frac{v^2 + a^2}{v^2}$$

$$\therefore dx = \frac{v^2}{a^2 + v^2} dv = \left(1 - \frac{a^2}{a^2 + v^2}\right) dv.$$

$$\text{Integrating, } x + C = v - a \tan^{-1} \frac{v}{a}$$

$$\text{or } x + C = (x + y) - a \tan^{-1} \frac{x + y}{a}$$

$$\text{or } y = C + a \tan^{-1} \frac{x + y}{a} \text{ is the solution.}$$

Ex. 5. Solve $\frac{x dx + y dy}{x dy - y dx} = \sqrt{\frac{a^2 - x^2 - y^2}{x^2 + y^2}}$.

[Delhi Hons 62, Agra B. Se. 55]

Solution. Here we change to polar co-ordinates by putting

$$x = r \cos \theta, y = r \sin \theta, x^2 + y^2 = r^2, x dx + y dy = r dr.$$

$$\frac{y}{x} = \tan \theta, \therefore \frac{x dy - y dx}{x^2} = \sec^2 \theta d\theta \text{ or } x dy - y dx = r^2 d\theta.$$

$$\therefore \text{the equation becomes } \frac{1}{r} \frac{dr}{d\theta} = \sqrt{\frac{a^2 - r^2}{r^2}}$$

$$\text{Separating the variables, } \frac{dr}{\sqrt{a^2 - r^2}} = d\theta.$$

$$\text{Integrating, } \sin^{-1}(r/a) = \theta + C \text{ or } r = a \sin(\theta + C).$$

$$\text{i.e., } \sqrt{x^2 + y^2} = a \sin[\tan^{-1}(y/x) + C].$$

Ex. 6. Solve $x \frac{dy}{dx} - y = x \sqrt{x^2 + y^2}$.

[Bombay 61; Agra 56]

Solution. The equation can be put as

$$x dy - y dx = x \sqrt{x^2 + y^2} dx \text{ or } \frac{x dy - y dx}{x^2} = \sec^2 \theta d\theta.$$

Changing to polars as above, the equation becomes

$$x^2 \sec^2 \theta d\theta = x r dx$$

$$\text{or } x \sec^2 \theta d\theta = r dx \text{ or } r \cos \theta \sec^2 \theta d\theta = r dx$$

$$\text{or } \sec \theta d\theta = dx, \text{ variables separated.}$$

$$\text{Integrating, } \log(\sec \theta + \tan \theta) = x + \log C.$$

$$\therefore \sec \theta + \tan \theta = ce^x \text{ or } \sqrt{1 + y^2/x^2} + y/x = ce^x.$$

Ex. 7. Solve $\left(\frac{x+y-a}{x+y-b}\right) \frac{dy}{dx} = \left(\frac{x+y+a}{x+y+b}\right)$

[Delhi Hons. 63; Nagpur 55]

$$\text{Solution. Put } x + y = v, \text{ so that } 1 + \frac{dy}{dx} = \frac{dv}{dx}.$$

$$\text{i.e., } \frac{dv}{dx} = 1 + \left(\frac{v+a}{v-b}\right) \left(\frac{v-b}{v-a}\right) = \frac{2(v^2 - ab)}{v^2 + (b-a)v - ab}$$

$$\text{or } 2 dx = \left(1 + \frac{b-a}{2} \frac{2v}{v^2 - ab}\right) dv.$$

Integrating, $2x + C = v + \frac{h-a}{2} \log(v^2 - ab)$

or $2x + C = x + y + \frac{1}{2}(h-a) \log[(x+y)^2 - ab]$ etc.

Ex. 8. $\frac{dy}{dx} = (x+y)^2$.

[Gauhati 62; Delhi 62; Raj. 62]

Hint. Put $x+y=c$ etc.

2.4. Homogeneous Differential Equations.

[Poona 61 (S)]

An equation of the form $\frac{dy}{dx} = \frac{f_1(x, y)}{f_2(x, y)}$ in which $f_1(x, y)$ and $f_2(x, y)$ are homogeneous functions* of x and y of the same degree can be reduced to an equation in which variables are separable by

putting $y = vx$, $\frac{dy}{dx} = v + x \frac{dv}{dx}$.

The following few examples will illustrate the method.

Ex. 1. Solve $(x^2 + y^2) dx + 2xy dy = 0$.

Solution. We have $\frac{dy}{dx} = -\frac{x^2 + y^2}{2xy}$ (homogeneous).

Putting $y = vx$, $\frac{dy}{dx} = v + x \frac{dv}{dx}$, the equation becomes

$$v + x \frac{dv}{dx} = -\frac{x^2 + x^2v^2}{2x \cdot vx} = -\frac{1 + v^2}{2v}$$

or $\frac{dv}{dx} = -\frac{1 + v^2}{2v} - v = -\frac{1 + 3v^2}{2v}$ (variable separable).

$$\therefore \frac{v}{1 + 3v^2} dv = -\frac{dx}{2x}$$

Integrating, $\log x + \frac{1}{2} \log(1 + 3v^2) = \log C$
or $x(1 + 3v^2)^{1/2} = C$ or $x(1 + 3y^2/x^2)^{1/2} = C$.

Ex. 2. Solve $x^2y dx - (x^3 + y^3) dy = 0$.

[Agra B Sc. 54]

Solution. We have $\frac{dy}{dx} = \frac{x^2y}{x^3 + y^3}$ (homogeneous).

Putting $y = vx$, $\frac{dy}{dx} = v + x \frac{dv}{dx}$, the equation becomes

$$v + x \frac{dv}{dx} = \frac{v}{1 + v^3} \quad \text{or} \quad x \frac{dv}{dx} = \frac{v}{1 + v^3} - v = -\frac{v^4}{1 + v^3}$$

$$\text{or} \quad \frac{dx}{x} = -\frac{1 + v^3}{v^4} dv = -\left[\frac{1}{v^4} + \frac{1}{v}\right] dv$$

Integrating, $\log x = \frac{1}{3v^3} - \log v + C$; $\log vx = \frac{1}{3v^3} + C$

*A function $f(x, y)$ is called homogeneous of degree n , if $f(\lambda x, \lambda y) = \lambda^n f(x, y)$.

or $\log y = \frac{x^3}{3y^3} + C$ as $v = \frac{y}{x}$.

Ex. 3. Solve $\frac{dy}{dx} = \frac{y^3 + 3x^2y}{x^3 + 3xy^2}$.

[Lucknow Pass 60]

Solution. Putting $y = vx$, $\frac{dy}{dx} = v + x \frac{dv}{dx}$, we get

$$x \frac{dv}{dx} = \frac{dy}{dx} - v = \frac{v^3 + 3v}{1 + 3v^2} - v = \frac{2v(1 - v^2)}{1 + 3v^2}$$

or $\frac{2 dx}{x} = \frac{1 + 3v^2}{2v(1 - v^2)} dv = \left(\frac{1}{v} - \frac{2}{1+v} + \frac{2}{1-v} \right) dv$.

Integrating,

$$2 \log x = \log v - 2 \log(1 - v) - 2 \log(1 + v) + \log C$$

or $x^2(1 - v)^2(1 + v)^2 = Cv$. Put $v = y/x$ etc.

Ex. 4. Solve $y^2 + x^2 \frac{dy}{dx} = xy \frac{dy}{dx}$.

[Delhi Hons. 66; Cal. Hons. 61, 56; Osmania 60; Gujrat 61]

Solution. The equation is $\frac{dy}{dx} = \frac{y^2}{xy - x^2}$ [homogeneous].

Putting $y = vx$, $\frac{dy}{dx} = v + x \frac{dv}{dx}$, we get

$$v + x \frac{dv}{dx} = \frac{v^2}{v - 1} \quad \text{or} \quad x \frac{dv}{dx} = \frac{v^2}{v - 1} - v$$

or $x \frac{dv}{dx} = \frac{v}{v - 1} \quad \text{or} \quad \frac{dx}{x} = \frac{v - 1}{v} dv$

or $\frac{dx}{x} = \left(1 - \frac{1}{v} \right) dv$.

Integrating, $\log x = v - \log v + \log c$

or $\log xv = v + \log c$ or $xv = ce^v$

or $y = ce^{y/x}$ as $y = vx$.

Ex. 5. Solve $(x^2 + y^2) dy = xy dx$

[Nau. ul. y D C 1961]

Hint. Homogeneous. Put $v = y/x$

Ans. $1 - (x^2 + y^2)$

Ex. 6. Solve the following homogeneous equations:

(i) $y(y^2 - 2x^2) dx + x(2y^2 - x^2) dy = 0$.

[Karnatak B Sc (Sub) 1960]

(ii) $\frac{1}{2x} \frac{dy}{dx} + \frac{x+y}{x^2+y^2} = 0$.

[Lucknow Pass 1955]

(iii) $\frac{dy}{dx} + \frac{y(x+y)}{x^2} = 0$.

Ans. $x^2y - c^2(y+2x)$

[Poona 1964; Nag 58; Kerala 61; Vikram 61]

(iv) $x^2y dx - x^3 dy = y^3 dy$.

Ans. $\log y = \frac{x^2}{3y^2} + C$.

$$(v) (x^2 - y^2) \frac{dy}{dx} = xy.$$

$$(vi) (x+y)^2 = xy \frac{dy}{dx}$$

[Poona 1964]

$$(vii) x \frac{dy}{dx} - y = \sqrt{(x^2 + y^2)}. \quad [\text{Sagar 1963; Cal. Hons. 62; Raj. 56}]$$

[Cf Ex. 6 P. 10]

$$\text{Ans. } x^2 + y^2 = (Cx^2 - y)^2.$$

$$\text{Ex. 7. } \left(x \cos \frac{y}{x} + y \sin \frac{y}{x} \right) y = \left(y \sin \frac{y}{x} - x \cos \frac{y}{x} \right) x \frac{dy}{dx}$$

[Cal. Hons 1962]

$$\text{or } x \cos \frac{y}{x} (y dx + x dy) = y \sin \frac{y}{x} (x dy - y dx).$$

[Raj. 1959; Cal. Hons. 61, 55; Delhi 68, 61]

$$\text{Solution. The equation is } \frac{dy}{dx} = \frac{y(y \sin y/x + x \cos y/x)}{x(y \sin y/x - x \cos y/x)}.$$

$$\text{Putting } y = vx, \quad \frac{dy}{dx} = \frac{dy}{dx} - v = \frac{2v \cos v}{v \sin v - \cos v}$$

$$\text{or } \left(\tan v - \frac{1}{v} \right) dv = 2 \frac{dx}{x}, \quad \text{i.e., } \log \frac{\sec v}{v} = \log C + 2 \log x$$

$$\text{or } v \cdot \sec(y/x) = Cxy \text{ is the solution.}$$

$$\text{Ex 8. Solve } \left(x \sin \frac{y}{x} \right) \frac{dy}{dx} = \left(y \sin \frac{y}{x} - x \right)$$

[Delhi Pass 67]

$$\text{Solution. Equation is } \frac{dy}{dx} = \frac{y}{x} - \operatorname{cosec} \frac{y}{x}$$

$$\text{Putting } y = vx, \quad \frac{dy}{dx} = x \frac{dv}{dx} + v.$$

$$\text{Equation reduces to } \sin v \, dv = -\frac{dx}{x}.$$

$$\text{Integrating, } -\cos v = -\log Cx$$

$$\text{or } \cos \frac{y}{x} = \log Cx \text{ is the solution.}$$

$$\text{Ex. 9. Solve } (x^2 + 2xy - y^2) dx + (y^2 + 2xy - x^2) dy = 0.$$

[Gujrat B.Sc. (Prin.) 1961]

$$\text{Solution. } \frac{dy}{dx} = -\frac{x^2 + 2xy - y^2}{y^2 + 2xy - x^2} \quad \text{Put } y = vx.$$

$$\therefore v + x \frac{dv}{dx} = -\frac{1 + 2v - v^2}{v^2 + 2v - 1}$$

$$x \frac{dv}{dx} = -\frac{1 + 2v - v^2}{v^2 + 2v - 1} - v = -\frac{v^3 + v^2 + v + 1}{v^2 + 2v - 1}$$

$$\therefore \frac{dx}{x} = -\frac{v^2 + 2v - 1}{v^3 + v^2 + v + 1} dv = -\frac{v^2 + 2v - 1}{(v+1)(v^2+1)} dv$$

$$= \left(\frac{1}{v+1} - \frac{2v}{v^2+1} \right) dv.$$

Integrating, $\log x = \log(v+1) - \log(v^2+1) + \log C$

or $\frac{x}{v^2+1} = C(v+1)$ or $\frac{x}{y^2/x^2+1} = C\left(\frac{y}{x}+1\right)$

Ex. 10. Solve $2y^3 dx + (x^2 - 3y^2) x dy = 0$.

[Bombay B.Sc. (Sub.) 1962]

Solution. Proceed yourself.

2.5. Equation Reducible to Homogeneous Form.

An equation of the type $\frac{dy}{dx} = \frac{ax+by+c}{a'x+b'y+c'}$, when $\frac{a}{a'} \neq \frac{b}{b'}$ can be reduced to homogeneous form as follows:

Put $x = X+h$, $y = Y+k$; then $\frac{dy}{dx} = \frac{dY}{dX}$, where X, Y are new variables and h, k are arbitrary constants. The equation now becomes

$$\frac{dY}{dX} = \frac{aX+bY+(ah+bk+c)}{a'X+b'Y+(a'h+b'k+c')}$$

We choose the constants h and k in such a way that $ah+bk+c=0$, $a'h+b'k+c'=0$.

With this substitution the differential equation reduces to $\frac{dY}{dX} = \frac{aX+bY}{a'X+b'Y}$ which is a homogeneous equation in X, Y and can be solved by putting $Y=vX$ as earlier.

[Special Case. When $\frac{a}{a'} = \frac{b}{b'} = \frac{1}{m}$ (say), then the differential equation can be written as

$$\frac{dy}{dx} = \frac{ax+by+c}{m(ax+by)+c}$$

Put $ax+by=v$, so that $a+b \frac{dy}{dx} = \frac{dv}{dx}$

(1) then becomes $\frac{1}{b} \left(\frac{dv}{dx} - a \right) = \frac{v+c}{mv+c}$ in which variables can be separated.

* Ex. 1. Solve $\frac{dy}{dx} = \frac{x+2y-3}{2x+y-3}$

Solution. Put $x = X+h$, $y = Y+k$, where h, k are some constants; then $\frac{dy}{dx} = \frac{dY}{dX}$. The given equation then becomes

$$\frac{dY}{dX} = \frac{X+2Y+(h+2k-3)}{2X+Y+(2h+k-3)}$$

[Vikram 60]

Now choose h, k such that $h+2k-3=0$ and $2h+k-3=0$. Solving these we get $h=1, k=1$.

$$\therefore \frac{dY}{dX} = \frac{X+2Y}{2X+Y} \text{ homogeneous in } X \text{ and } Y.$$

Put $Y=vX$, so that $\frac{dY}{dX} = v + X \frac{dv}{dX}$.

$$\therefore v + X \frac{dv}{dX} = \frac{X+2vX}{2X+vX} = \frac{1+2v}{2+v}, \text{ i.e., } X \frac{dv}{dX} = \frac{1+2v}{2+v} - v$$

$$\text{or } \frac{dX}{X} = \frac{2+v}{1-v^2} dv = \left(\frac{2}{1-v^2} + \frac{v}{1-v^2} \right) dv.$$

$$\text{Integrating, } \log X = 2 \cdot \frac{1}{2} \log \frac{1+v}{1-v} - \frac{1}{2} \log (1-v^2) + \log C$$

$$\text{or } X = C \frac{1+v}{1-v} \cdot \frac{1}{\sqrt{(1-v^2)}} = \frac{C\sqrt{1+v}}{(1-v)^{3/2}}$$

$$\text{or } X^2(1-v)^3 = C^2(1+v)$$

$$\text{or } X^2 \left(1 - \frac{Y}{X}\right)^3 = C^2 \left(1 + \frac{Y}{X}\right) \text{ as } v = \frac{Y}{X}$$

$$\text{or } (X-Y)^3 = C^2(X+Y) \text{ but } x = X+1, y = Y+1.$$

$$\therefore (x-y)^3 = C^2(x+y-2) \text{ is the required solution.}$$

Ex. 2. Solve $(3x-7y-3) \frac{dy}{dx} = 3y-7x+7$.

[Raj. M.Sc. 61]

Solution. $\frac{dy}{dx} = \frac{3y-7x+7}{3x-7y-3}$

Put $x = X+h, y = Y+k$, where h, k are some constants. Then $\frac{dy}{dx} = \frac{dY}{dX}$. And the given equation becomes

$$\frac{dY}{dX} = \frac{3Y-7X+(3k-7h+7)}{3X-7Y+(3h-7k-3)}$$

Choose h, k such that $3h-7k-3=0$ and $3k-7h+7=0$, which give $h=1, k=0$.

$$\therefore \frac{dY}{dX} = \frac{3Y-7X}{3X-7Y} \text{ (homogeneous).}$$

Put $Y=vX, \frac{dY}{dX} = v + X \frac{dv}{dX}$

$$\therefore v + X \frac{dv}{dX} = \frac{3vX-7X}{3X-7vX} = \frac{3v-7}{3-7v}$$

$$\text{or } X \frac{dv}{dX} = \frac{3v-7}{3-7v} - v = \frac{7(v^2-1)}{3-7v}$$

$$\text{or } \frac{7}{X} dX = \frac{3-7v}{(v^2-1)} dv = - \left(\frac{2}{v-1} + \frac{5}{v+1} \right) dv.$$

Integrating, $7 \log X = -2 \log (v-1) - 5 \log (v+1) + \log C$
 or $X^7 (v-1)^2 (v+1)^5 = C$
 or $X^7 \left(\frac{Y}{X} - 1\right)^2 \left(\frac{Y}{X} + 1\right)^5 = C$ as $Y = vX$
 or $(Y-X)^2 (Y+X)^5 = C$
 or $(y-x+1)^2 (y+x-1)^5 = C$ as $x = X+1, y = Y+0$.

Ex. 3. Solve $(2x+y+3) \frac{dy}{dx} = x+2y+3$.

[Karnatak B.Sc. (Princ.) 60]

Solution. $\frac{dy}{dx} = \frac{x+2y+3}{2x+y+3}$.

Put $x = X+h, y = Y+k$, where h, k are constants.

$$dx = dX, \quad dy = dY; \quad \therefore \frac{dy}{dx} = \frac{dY}{dX}$$

$$\therefore \frac{dY}{dX} = \frac{X+2Y+(h+2k+3)}{2X+Y+(2h+k+3)}$$

Choose h, k such that $h+2k+3=0, 2h+k+3=0$. Solving these, we get $h=-1, k=-1$.

$$\therefore \frac{dY}{dX} = \frac{X+2Y}{2X+Y}. \quad \text{Put } Y = vX, \quad \frac{dY}{dX} = v + X \frac{dv}{dX}.$$

$$\therefore v + X \frac{dv}{dX} = \frac{X+2vX}{2X+vX} \quad \text{or} \quad X \frac{dv}{dX} = \frac{1+2v}{2+v} - v$$

$$\text{or} \quad \frac{dX}{X} = \frac{2+v}{1-v^2} dv = \left(\frac{\frac{3}{2}}{1-v} + \frac{\frac{1}{2}}{1+v} \right) dv.$$

Integrating, $2 \log X = -3 \log (1-v) + \log (1+v) + \log C$

$$\text{or} \quad X^2 \frac{(1-v)^3}{1+v} = C \quad \text{or} \quad X^2 \frac{(1-Y/X)^3}{(1+Y/X)} = C$$

$$\text{or} \quad (X-Y)^3 = C(X+Y); \quad \text{where } x = X-1, y = Y-1$$

$$\text{or} \quad (X-y)^3 = C(x+y-2) \text{ is the solution.}$$

Ex. 4. Solve $(2x-2y+5) \frac{dy}{dx} = x-y+3$.

[Sagar 63; Agra B.Sc. 61, 52]

Solution. The equation is $\frac{dy}{dx} = \frac{x-y+3}{2(x-y)+5}$.

$$\text{Put } x-y=v, \text{ so that } 1 - \frac{dy}{dx} = \frac{dv}{dx} \quad \text{or} \quad \frac{dy}{dx} = 1 - \frac{dv}{dx}.$$

$\therefore$ The equation becomes

$$1 - \frac{dv}{dx} = \frac{v+3}{2v+5} \quad \text{or} \quad \frac{dv}{dx} = 1 - \frac{v+3}{2v+5} = \frac{v+2}{2v+5}$$

$$\text{or} \quad dx = \frac{2v+5}{v+2} dv = \left(2 + \frac{1}{v+2} \right) dv, \text{ separating the variables.}$$

Integrating. $x = 2v + \log(v+2) + C$,

$x = 2(x-y) + \log(x-y+2) + C$ as $v = x-y$

or $2y - x = \log(x-y+2) + C$ is the required solution.

Ex. 5 Solve $\frac{dy}{dx} = \frac{6x-4y+3}{3x-2y+1}$

[Poona 64; Karnatak B.Sc. (Princ.) 61]

Solution. Put $3x-2y=v$, i.e., $3-2\frac{dy}{dx} = \frac{dv}{dx}$

$$\therefore \frac{dv}{dx} = 3 - 2\frac{2v+3}{v+1} = -\frac{v+3}{v+1}$$

$$\therefore dx = -\frac{v+1}{v+3} dv = -\left(1 - \frac{2}{v+3}\right) dv$$

Integrating, $x = -v + 2 \log(v+3) + C$

or $x = (2y-3x) + 2 \log(3x-2y+3) + C$

or $2x-y = \log(3x-2y+3) + \frac{1}{2}C$ is the solution.

Ex. 6 Solve $(5x-4y+1)\frac{dy}{dx} = (3x-2y+1)$.

[Karnatak B.Sc. (Sub.) 61]

Solution. $\frac{dy}{dx} = \frac{3x-2y+1}{2(3x-2y)+1}$ Put $3x-2y=v$.

$$\frac{dv}{dx} = 3 - 2\frac{dv}{dx} = 3 - 2\frac{v+1}{2v+1} = \frac{4v+1}{2v+1}$$

$$dx = \frac{2v+1}{4v+1} dv \text{ or } 2 dx = \left(1 + \frac{1}{4v+1}\right) dv \text{ etc.}$$

Ex. 7 Solve the following equations :

(i) $(2x+y+1) dx + (4x+2y-1) dy = 0$.

[Gujrat B.Sc. (Princ.) 61]

(ii) $\frac{dy}{dx} = \frac{6x-2y-7}{3x-y+4}$

[Luck. Pass 56]

(iii) $(2x-5y+3) dx - (2x+4y-6) dy = 0$. [Delhi Hons. 61]

(iv) $\frac{dy}{dx} = \frac{y-x+1}{y-x-5}$

[Poona 62; Nag. 62]

(v) $\frac{dy}{dx} = \frac{3x-4y-2}{2x-4y-3}$

[Cal. Hons 63]

(vi) $(3y+2x+4) dx - (4x+6y+5) dy = 0$. [Karnatak 63]

(vii) $(2x-5y+3) dx - (2x+4y-6) dy = 0$. [Delhi Hons. 65]

(viii) $(x-y-2) dx + (x-2y-3) dy = 0$. [All. 66]

(ix) $(4x+2y+1) dy = (2x+y+3) dx$. [Delhi Pass 67]

Hint. In (i) put $2x+y=v$, in (ii) put $3x-y=v$ and (iii) can be reduced to homogeneous form as usual. In (ix) putting $v=2x+y$, variables can be separated.

Ex. 8. Solve $2y \frac{dy}{dx} = \frac{x+y^2}{x+4y^2}$ [Bombay B.Sc. 61]

Solution. Put $y^2 = v$, $2y \frac{dy}{dx} = \frac{dv}{dx}$.

$\therefore \frac{dv}{dx} = \frac{x+v}{x+4v}$ [homogeneous]. Now put $v = xz$ etc.

2.6. A particular case

A differential equation of the form

$$\frac{dy}{dx} = \frac{ax+by+c}{-bx+hy+k'}$$

in which coefficient of y in the numerator is equal to the coefficient of x in the denominator with sign changed, can be integrated as follows :

The equation (1) can be written as

$$-b(x dy + y dx) + (hy + k) dy - (ax + c) dx = 0.$$

Integrating, we get $-bxy + (\frac{1}{2}hy^2 + ky) - (\frac{1}{2}ax^2 + cx) = A$.

Ex. 1. Solve $\frac{dy}{dx} + \frac{ax+hy+g}{hx+by+f} = 0$.

[Raj. B.Sc. 66; Agra B.Sc. 57; Delhi B.A. 57; Raj. M.Sc. 62]

Solution. The equation can be written as

$$(hx+by+f) dy + (ax+hy+g) dx = 0$$

or $h(x dy + y dx) + (by+f) dy + (ax+g) dx = 0$.

Integrating, $hxy + \frac{1}{2}by^2 + fy + \frac{1}{2}ax^2 + gx = A$

or $ax^2 + 2hxy + by^2 + 2fy + 2gx + c = 0$, writing $c = -2A$.

Ex. 2. Solve $\frac{dy}{dx} = \frac{2x-y+1}{x+2y-3}$ [Agra B.Sc. 59; Nag. 53 (S)]

Solution. Here coefficient of y in numerator is equal to coefficient of x in the denominator with sign changed. Hence write it as

$$(x+2y-3) dy - (2x-y+1) dx = 0$$

or $(x dy + y dx) + (2y-3) dy - (2x+1) dx = 0$.

Integrating, $xy + y^2 - 3y - x^2 - x = C$.

Ex. 3. Solve $(2x-y+1) dx + (2y-x-1) dy = 0$.

[Bombay B.Sc. (Sub.) 61; Poona 61]

Solution. The equation is of above type. Hence after regrouping, we have

$$(2x+1) dx + (2y-1) dy - (y dx + x dy) = 0.$$

Integrating, $(x^2+x) + (y^2-y) - xy = C$,

which is the solution.

Ex. 4. Solve $\frac{dy}{dx} + \frac{2x+3y+1}{3x+4y-1} = 0$.

[Delhi Hons. 60]

Solution. The equation is of the above type and can be written as
 $(3x+4y-1) dy + (2x+3y+1) dx = 0$,
 i.e., $3(x dy + y dx) + (4y-1) dy + (2x+1) dx = 0$.

Integrating, $3xy + 2y^2 - y + x^2 + x = C$ is the solution.

2.7. Linear Differential Equations

[Poona 63, 61; Nagpur 62, 61; Guj 61]

A differential equation of the form

$$\frac{dy}{dx} + Py = Q,$$

where P, Q are functions of x or constants, is called the linear differential equation of the first order.

To solve this equation, multiply both the sides by $e^{\int P dx}$

Then it becomes $e^{\int P dx} \frac{dy}{dx} + Py e^{\int P dx} = Q e^{\int P dx}$.

or $\frac{d}{dx} [y e^{\int P dx}] = Q e^{\int P dx}$.

Integrating both the sides, w.r.t. x , we get

$$y e^{\int P dx} = \int [Q e^{\int P dx}] dx + C,$$

which is the required solution.

Integrating factor (I.F.). It will be noticed that for solving (1), we multiplied it by a factor $e^{\int P dx}$ and the equation became readily (directly) integrable. Such a factor is called the integrating factor.

Note. Sometimes a differential equation takes linear form if we regard x as dependent variable and y as independent variable.

The equation can then be put as $\frac{dx}{dy} + Px = Q$, where P, Q are functions of y or constants.

The integrating factor in this case is $e^{\int P dy}$ and solution is

$$x e^{\int P dy} = \int [Q e^{\int P dy}] dy + C.$$

(See Ex. 1 to 4 pages 21 and 22).

Ex. 1. Solve $(1-x^2) \frac{dy}{dx} - xy = 1$.

[Delhi 68 : Nag. 61]

Solution. The equation can be written as

$$\frac{dy}{dx} - \frac{x}{1-x^2} y = \frac{1}{1-x^2}$$

This is now expressed in the linear form

$$P = -\frac{x}{1-x^2}, \quad \text{I.F.} = e^{\int P dx} = e^{\int \frac{-x}{1-x^2} dx} = e^{-\frac{1}{2} \log(1-x^2)} \\ = \sqrt{1-x^2}.$$

Hence the solution is

$$y \cdot \sqrt{1-x^2} = \int \frac{1}{\sqrt{1-x^2}} \sqrt{1-x^2} dx + C.$$

Ex. 2. (a) Solve $x \frac{dy}{dx} + 2y = x^2 \log x$. [Lucknow 52]

Solution. The equation is $\frac{dy}{dx} + \frac{2}{x} y = x \log x$.

$$\text{I.F.} = e^{\int (2/x) dx} = e^{2 \log x} = x^2.$$

Hence the solution is

$$y \cdot x^2 = C + \int x^2 \cdot x \log x dx = C + \int x^3 \log x dx \\ = C + \log x \cdot \frac{x^4}{4} - \int \frac{1}{x} \cdot \frac{x^4}{4} dx \\ = C + \frac{1}{4} x^4 \log x - \frac{1}{16} x^4 \\ \text{or } y = Cx^{-2} + \frac{1}{4} x^2 (\log x - \frac{1}{4}).$$

Ex. 2. (b) Solve $x \frac{dy}{dx} + 2y = x^4$.

[Bombay B.Sc. 61]

Solution. Equation is $\frac{dy}{dx} + \frac{2}{x} y = x^3$. I.F. = x^2 as above.

$$\text{Solution is } y \cdot x^2 = C + \int x^3 \cdot x^2 dx = C + \frac{1}{6} x^6.$$

* Ex. 3) Solve $(x^3 - x) \frac{dy}{dx} - (3x^2 - 1) y = x^5 - 2x^3 + x$.

[Gujrat B.Sc. (Sub.) 1961]

Solution. The equation is

$$\frac{dy}{dx} - \frac{3x^2-1}{x^3-x} y = (x^2-1).$$

$$\text{I.F.} = e^{-\int \frac{(3x^2-1)}{x^3-x} dx} = e^{-\log(x^3-x)} = \frac{1}{x^3-x}.$$

$$\therefore \text{Solution is } y \cdot \frac{1}{x^3-x} = C + \int \frac{x^2-1}{x^3-x} dx \\ = C + \int \frac{1}{x} dx = C + \log x.$$

Ex 4. Solve $xp + y = ax^2 + bx + c, p = \frac{dy}{dx}$.

[Delhi Hons. 1957]

Solution. The equation can be written as

$$\frac{dy}{dx} + \frac{1}{x} y = ax + b + \frac{c}{x} \text{ [linear].}$$

$$\text{I.F.} = e^{\int \frac{1}{x} dx} = e^{\log x} = x.$$

$$\begin{aligned} \therefore y \cdot x &= C + \int \left(ax + b + \frac{c}{x} \right) x dx = C + \int (ax^2 + bx + c) dx \\ &= C + \frac{1}{3} ax^3 + \frac{1}{2} bx^2 + cx. \end{aligned}$$

✓ Ex. 5. If $\frac{dy}{dx} + 2y \tan x = \sin x$ and if $y=0$ when $x=\frac{1}{2}\pi$, express y in terms of x . [Poona 1964 ; Nagpur 61]

Solution. The equation is linear.

$$\text{I.F.} = e^{\int 2 \tan x dx} = e^{-2 \log \cos x} = \sec^2 x.$$

Hence general solution is

$$y \cdot \sec^2 x = C + \int \sin x \sec^2 x dx = C + \int \sec x \tan x dx$$

$$\text{or } y \sec^2 x = C + \sec x.$$

When $y=0$, $x=\frac{1}{2}\pi$, $\therefore 0 = C + \sec \frac{1}{2}\pi$ or $C+2=0$, $C=-2$.

$$\begin{aligned} \text{Hence solution is } y \sec^2 x &= \sec x - 2, \\ y &= \cos x - 2 \cos^3 x. \end{aligned}$$

Ex. 6. Solve $x(x-1) \frac{dy}{dx} - y = x^2(x-1)^2$. [Luck. Pass 1958]

Solution. Equation is $\frac{dy}{dx} - \frac{1}{x(x-1)} y = x(x-1)$.

$$\text{I.F.} = e^{-\int \frac{1}{x(x-1)} dx} = e^{\int \left(\frac{1}{x} - \frac{1}{x-1} \right) dx} = \frac{x}{x-1}.$$

$$\text{Hence } y \cdot \frac{x}{x-1} = C + \int x(x-1) \cdot \frac{x}{x-1} dx = C + \int x^2 dx$$

$$\text{or } y \cdot \frac{x}{x-1} = C + \frac{1}{3} x^3.$$

Ex. 7. Solve $(1+x) \frac{dy}{dx} + 3y = \frac{1+x+x^2}{(1+x)}$. [Lucknow Pass 1957]

Solution. Equation is $\frac{dy}{dx} + \frac{3}{1+x} y = \frac{1+x+x^2}{(1+x)^2}$.

$$\text{I.F.} = e^{\int \frac{3}{1+x} dx} = e^{3 \log(1+x)} = (1+x)^3$$

$$\therefore y(1+x)^3 = C + \int \frac{(1+x+x^2)}{(1+x)^4} (1+x)^3 dx$$

$$= C + \int \frac{1+x+x^2}{1+x} = C + \int \left(\frac{1}{1+x} + x \right) dx$$

$$= C + \log(1+x) + \frac{1}{2}x^2.$$

Ex. 8. Solve $x \frac{dy}{dx} + 2y = \frac{dy}{dx} + 4$. [Nagpur T.D.C. 1961 (S)]

Solution. The equation can be written as

$$(x-1) \frac{dy}{dx} + 2y = 4 \quad \text{or} \quad \frac{dy}{dx} + \frac{2}{x-1} y = \frac{4}{x-1}.$$

Linear, I.F. = $e^{\int \frac{2}{x-1} dx} = e^{2 \log(x-1)} = (x-1)^2.$

$$y(x-1)^2 = \int \frac{4}{x-1} (x-1)^2 dx + C$$

$$y(x-1)^2 = 2(x-1)^2 + C, \text{ which is the solution.}$$

Ex. 9. Solve $x \frac{dy}{dx} - 2y = x^2 + \sin \frac{1}{x^2}.$

[Bombay B.A. (Sub.) 1958]

Solution. The equation is $\frac{dy}{dx} - \frac{2}{x}y = x + \frac{1}{x} \sin \frac{1}{x^2}.$

$$\text{I.F.} = e^{-\int \frac{2}{x} dx} = e^{-2 \log x} = \frac{1}{x^2}.$$

$$\therefore y \cdot \frac{1}{x^2} = C + \int x \frac{1}{x^2} dx + \int \frac{1}{x^3} \sin \frac{1}{x^2} dx.$$

$$= C + \log x - \frac{1}{2} \int \sin t dt, \text{ where } \frac{1}{x^2} = t, \quad \frac{-2}{x^3} dx = dt$$

$$= C + \log x + \frac{1}{2} \cos t$$

$$= C + \log x + \frac{1}{2} \cos \frac{1}{x^2}.$$

Ex. 10. Solve $\frac{dy}{dx} - 2y \cos x = -2 \sin 2x.$

[Vikram 65; Gujrat B.Sc. (Sub.) 61]

Solution. I.F. = $e^{-2 \int \cos x dx} = e^{-2 \sin x}.$

$\therefore$ Solution is

$$ye^{-2 \sin x} = C - 2 \int \sin 2x e^{-2 \sin x} dx$$

$$= C - 4 \int \sin x \cos x e^{-2 \sin x} dx; \text{ put } -2 \sin x = t$$

$$= C - \int t e^t dt = C - e^t (t - 1).$$

$$\therefore y = Ce^{2 \sin x} + (2 \sin x + 1) \text{ is the solution.}$$

Equations which become linear when x is treated as dependent variable.

Ex. 11. Solve $y \log y dx + (x - \log y) dy = 0.$

[Poona T.D.C. 61(S)]

Solution. Write the equation as

$$\frac{dx}{dy} + \frac{1}{y \log y} x = \frac{1}{y}.$$

$$\text{I.F.} = e^{\int \frac{1}{y \log y} dy} = e^{\log(\log y)} = \log y.$$

$$\begin{aligned} \star \quad x \log y &= C + \int \frac{1}{y} \log y \, dy \\ &= C + \frac{1}{2} (\log y)^2 \text{ is the solution.} \end{aligned}$$

Ex 2. Solve $dx + x \, dy = e^{-y} \log y \, dy$.

[Poona 61]

Solution. The equation can be written as

$$\frac{dx}{dy} + x = e^{-y} \log y, \text{ I.F.} = e^y.$$

$$\begin{aligned} \therefore x e^y &= C + \int e^{-y} \log y \cdot e^y \, dy \\ &= C + \int \log y \, dy = C + \log y \cdot y - \int y \cdot \frac{1}{y} \, dy \\ &= C + y \log y - y. \end{aligned}$$

Ex. 3. Solve $(1+y^2) \, dx + (x - \tan^{-1} y) \, dy = 0$.

[Gujrat 65;

Delhi Hons. 65 ; Pb. 62; Cal. Hons. 62; Agra 67, 58]

Solution. The equation can be written as

$$\frac{dx}{dy} + \frac{1}{1+y^2} x = \frac{\tan^{-1} y}{1+y^2}.$$

$$\text{I.F.} = e^{\int \frac{1}{1+y^2} dy} = e^{\tan^{-1} y}.$$

$$\begin{aligned} \therefore x e^{\tan^{-1} y} &= \int \frac{\tan^{-1} y}{1+y^2} e^{\tan^{-1} y} \, dy + C \\ &= \int t e^t \, dt + C \text{ where } t = \tan^{-1} y \\ &= e^t (t-1) + C = e^{\tan^{-1} y} (\tan^{-1} y - 1) + C. \end{aligned}$$

Hence $x = (\tan^{-1} y - 1) + C e^{-\tan^{-1} y}$ is the solution.

Ex. 4. Solve $(x+2y^2) \frac{dy}{dx} = y$.

[Agra B.Sc. 1956 ; Raj B.Sc. 56]

Hint. The equation can be written as

$$\frac{dx}{dy} = x + 2y^2 \text{ [linear].}$$

$$\text{Ans. } x = y^2 + Cy.$$

2.8. Equations reducible to linear form

Bernoulli Equation*. $\frac{dy}{dx} + Py = Qy^n$.

*Known after James Bernoulli. The method of solution was discovered by Leibnitz.

where P and Q are functions of x or constants.

[Nag. T.D.C. 1961; Poona T.D.C. 61; Gujrat B.Sc. (Prin.) 58;
Poona B.A. (Gen.) 60]

Dividing both the sides by y^n we have

$$y^{-n} \frac{dy}{dx} + P y^{-n+1} = Q. \quad \dots(1)$$

Now put $y^{-n+1} = v$ so that $(1-n) y^{-n} \frac{dy}{dx} = \frac{dv}{dx}$.

Then (1) becomes $\frac{1}{1-n} \frac{dv}{dx} + Pv = Q$

or $\frac{dv}{dx} + P(1-n)v = (1-n)Q$

which is a linear equation in v and x .

II. Equation $f'(y) \frac{dy}{dx} + Pf(y) = Q$,

where P and Q are functions of x or constants.

Put $f(y) = v$ so that $f'(y) \frac{dy}{dx} = \frac{dv}{dx}$.

$\therefore$ equation becomes $\frac{dv}{dx} + Pv = Q$,

which is a linear equation in v and x .

Note. In each of these equations, single out Q (function on the right) and then make suitable substitution to reduce the equation in linear form.

* Ex. 12 Solve $\frac{dy}{dx} = x^3 y^3 - xy$.

[Karnatak B.Sc. (Prin.) 1960, 62; Agra 61; Bihar 62;
Gujrat B.Sc. (Sub.) 61]

Solution. The equation is $\frac{dy}{dx} + xy = x^3 y^3$.

Dividing by y^3 ; $\frac{1}{y^3} \frac{dy}{dx} + x \cdot \frac{1}{y^2} = x^3$.

Put $\frac{1}{y^2} = v$, so that $-\frac{2}{y^3} \frac{dy}{dx} = \frac{dv}{dx}$, i.e., $\frac{1}{y^3} \frac{dy}{dx} = -\frac{1}{2} \frac{dv}{dx}$.

$\therefore$ equation becomes $-\frac{1}{2} \frac{dv}{dx} + xv = x^3$

or $\frac{dv}{dx} - 2x \cdot v = -2x^3$.

Linear, I. F. = $e^{\int -2x dx} = e^{-x^2}$.

Hence $ve^{-x^2} = \int -2x^3 e^{-x^2} dx + C$
 $= \int x^2 (-2x) e^{-x^2} dx + C$

$$= \int -te^t dt + C \text{ where } t = -x^2$$

$$= -te^t + e^t + C = -e^{-x^2}(x^2 - 1) + C$$

Hence $v = 1 - x^2 + Ce^{x^2}$ or $\frac{1}{y^2} = 1 - x^2 + Ce^{x^2}$.

Ex. 2. Solve $\frac{dy}{dx} + xy = xy^2$.

[Karnatak 1960]

Solution. Dividing by y^2 , $y^{-2} \frac{dy}{dx} + xy^{-1} = x$.

Put $y^{-1} = v$, so that $-y^2 \frac{dy}{dx} = \frac{dv}{dx}$.

$\therefore$ equation is $\frac{dv}{dx} - xv = -x$.

I. F. $= e^{\int -x dx} = e^{-\frac{1}{2}x^2}$.

$\therefore ve^{-\frac{1}{2}x^2} = C - \int xe^{-\frac{1}{2}x^2} dx$

$= C + \int e^t dt$, where $-\frac{1}{2}x^2 = t$, $-x dx = dt$

or $y^{-1}e^{-\frac{1}{2}x^2} = C + e^t = C + e^{-\frac{1}{2}x^2}$

or $y^{-1} = Ce^{\frac{1}{2}x^2} + 1$ is the solution.

Ex 3. Solve $\frac{dy}{dx} + \frac{2}{x}y = \frac{y^3}{x^3}$.

[Nag. 1958]

Solution. Dividing by y^3 , $y^{-3} \frac{dy}{dx} + \frac{2}{x}y^{-2} = \frac{1}{x^3}$.

Put $y^{-2} = v$, so that $-2y^{-3} \frac{dy}{dx} = \frac{dv}{dx}$.

$\therefore$ equation becomes $-\frac{1}{2} \frac{dv}{dx} + \frac{2}{x}v = \frac{1}{x^3}$

or $\frac{dv}{dx} - \frac{4}{x}v = -\frac{2}{x^3}$.

I. F. $= e^{\int (-4/x) dx} = e^{-4 \log x} = \frac{1}{x^4}$.

$\therefore v \frac{1}{x^4} = \int -\frac{2}{x^3} \cdot \frac{1}{x^4} dx + C = C + \frac{1}{3x^3}$

or $\frac{1}{y^2} \cdot \frac{1}{x^4} = \frac{1}{3x^3} + C$ is the solution.

Ex. 4. Solve $\frac{dy}{dx} (x^2y^3 + xy) = 1$.

[Sagar 1962; Raj. 63; Cal. Hons. 62; Luck. 63]

Solution. The equation can be written as

$$\frac{dx}{dy} - xy = x^2y^3.$$

Dividing by x^2 , $x^{-2} \frac{dx}{dy} - \frac{1}{x} y = y^3$.

Put $-\frac{1}{x} = v$; $\therefore x^{-2} \frac{dx}{dy} = \frac{dv}{dy}$.

$\therefore$ equation becomes $\frac{dv}{dy} + vy = y^3$.

Linear in v and y . I. F. $= e^{\int y dy} = e^{\frac{1}{2}y^2}$.

$$\begin{aligned} \therefore v e^{\frac{1}{2}y^2} &= \int y^3 e^{\frac{1}{2}y^2} dy + C, \quad \frac{1}{2}y^2 = t, \quad y dy = dt \\ &= 2 \int t e^t dt + C = 2e^t (t-1) + C \end{aligned}$$

$$\text{or } -\frac{1}{x} e^{\frac{1}{2}y^2} = 2e^{\frac{1}{2}y^2} (\frac{1}{2}y^2 - 1) + C$$

or $\frac{1}{x} = (2 - y^2) - C e^{-\frac{1}{2}y^2}$ is the solution.

Ex. 5. Solve $\frac{dy}{dx} = 1 - x(y-x) - x^2(y-x)^2$.

[Karnatak 1961]

Solution. Put $y-x=v$, $\frac{dy}{dx} - 1 = \frac{dv}{dx}$.

$\therefore$ equation is $\frac{dv}{dx} + 1 = 1 - xv - x^2v^2$

$$\text{or } \frac{dv}{dx} + xv = -x^2v^2 \quad \text{or } v^{-2} \frac{dv}{dx} + xv^{-2} = -x^2.$$

Put $v^{-2} = u$, $-2v^{-3} \frac{dv}{dx} = \frac{du}{dx}$.

$\therefore$ the equation is $-\frac{1}{2} \frac{du}{dx} + xu = -x^2$

$$\text{or } \frac{du}{dx} - 2xu = 2x^2.$$

Linear in u and x . I. F. $= e^{\int -2x dx} = e^{-x^2}$.

$$\begin{aligned} \therefore u e^{-x^2} &= \int 2x^2 e^{-x^2} + C \quad -x^2 = t, \quad -2x dx = dt \\ &= \int t e^t dt + C = e^t (t-1) + C \end{aligned}$$

$$\text{or } v^{-2} e^{-x^2} = e^{-x^2} (-x^2 - 1) + C$$

or $(y-x)^{-2} = C e^{x^2} - (1+x^2)$ is the solution.

Ex. 6. Solve $2 \frac{dy}{dx} - \frac{y}{x} = \frac{y^3}{x^2}$.

[Nagpur 1961; Nagpur 51; Delhi Pass 57]

Solution. The equation is $2y^{-2} \frac{dy}{dx} - \frac{1}{x} y^{-1} = \frac{1}{x^2}$.

Put $-y^{-1} = v$, $\therefore y^{-2} \frac{dy}{dx} = \frac{dv}{dx}$.

$\therefore 2 \frac{dv}{dx} + \frac{1}{x} v = \frac{1}{x^2}$ or $\frac{dv}{dx} + \frac{1}{2x} v = \frac{1}{2x^2}$

Linear, I. F. $= e^{\int \frac{1}{2} \cdot \frac{1}{x} dx} = e^{\frac{1}{2} \log x} = \sqrt{x}$.

$\therefore v\sqrt{x} = C + \int \frac{1}{2x^2} \sqrt{x} dx = C + \frac{1}{2} \int x^{-3/2} dx$.

$\therefore -y^{-1} \sqrt{x} = C - x^{-1/2}$

or $\frac{1}{y} = -Cx^{-1/2} + x^{-1}$, is the solution.

Ex. 7. Solve $x \frac{dy}{dx} + y = y^2 \log x$.

[Luck. 1956]

Solution. Dividing by xy^2 , $y^{-2} \frac{dy}{dx} + \frac{1}{x} y^{-1} = \frac{1}{x} \log x$.

Put $-y^{-1} = v$, $y^{-2} \frac{dy}{dx} = \frac{dv}{dx}$.

$\therefore \frac{dv}{dx} - \frac{1}{x} v = \frac{1}{x} \log x$. I. F. $= e^{\int -\frac{1}{x} dx} = \frac{1}{x}$.

Hence solution is $v \cdot \frac{1}{x} = C + \int \frac{1}{x} \log x dx$

or $-\frac{1}{y} \cdot \frac{1}{x} = C + \log x \left(-\frac{1}{x}\right) - \int \frac{1}{x} \cdot \left(-\frac{1}{x}\right) dx$

integrating by parts

or $-\frac{1}{xy} = C - \frac{1}{x} \log x - \frac{1}{x}$

or $\frac{1}{y} = (1 + \log x) - Cx$ is the solution.

Ex 8. Solve $(x - y^2) dx + 2xy dy = 0$.

[Poona 1961]

Solution. The equation is

$2y \frac{dy}{dx} - \frac{1}{x} y^2 = -1$.

Put $y^2 = v$, $2y \frac{dy}{dx} = \frac{dv}{dx}$, $\therefore$ equation is $\frac{dv}{dx} - \frac{1}{x} v = -1$

I. F. $= e^{\int -\frac{1}{x} dx} = e^{-\log x} = \frac{1}{x}$.

$\therefore v \cdot \frac{1}{x} = C - \int \frac{1}{x} dx$ or $\frac{y^2}{x} = C - \log x$.

Ex. 9. (a) Solve $(x^2 + 3x + 2) \frac{dy}{dx} + (2x + 1)y = (xy + 2y)^2$.

[Bombay B.Sc. (Prin.) 1961]

Solution. The equation can be written as

$$(x+2)(x+1) \frac{dy}{dx} + (2x+1)y = y^2(x+2)^2$$

$$\text{or } y^{-2} \frac{dy}{dx} + \frac{2x+1}{(x+2)(x+1)} y^{-1} = \frac{x+2}{x+1}$$

[dividing by $y^2(x+2)(x+1)$].

Put $-y^{-1} = v$, $y^{-2} \frac{dy}{dx} = \frac{dv}{dx}$.

$\therefore$ equation is $\frac{dv}{dx} - \frac{2x+1}{(x+2)(x+1)} v = \frac{x+2}{x+1}$.

This is a linear equation.

$$\begin{aligned} \text{I.F.} &= e^{-\int \frac{2x+1}{(x+2)(x+1)} dx} = e^{\int \left(\frac{3}{x+2} - \frac{1}{x+1} \right) dx} \\ &= e^{\log(x+1) - 3 \log(x+2)} = \frac{x+1}{(x+2)^3} \end{aligned}$$

Hence $v \cdot \frac{x+1}{(x+2)^3} = C + \int \frac{x+2}{x+1} \cdot \frac{x+1}{(x+2)^3} dx$

or $-\frac{1}{y} \cdot \frac{x+1}{(x+2)^3} = C + \int \frac{1}{(x+2)^2} dx = C - \frac{1}{x+2}$.

Ex. 10. $\frac{dy}{dx} - 2y \tan x = y^2 \tan^2 x$.

[Bombay 61]

Solution. Dividing by y^2 , the equation becomes

$$y^{-2} \frac{dy}{dx} - 2y^{-1} \tan x = \tan^2 x.$$

Put $-y^{-1} = v$, $y^{-2} \frac{dy}{dx} = \frac{dv}{dx}$.

$\therefore$ equation is $\frac{dv}{dx} + 2 \tan x \cdot v = \tan^2 x$, I.F. = $\sec^2 x$.

$\therefore v \sec^2 x = C + \int \tan^2 x \cdot \sec^2 x dx = C + \frac{1}{3} \tan^3 x$

or $-\frac{1}{y} \sec^2 x = C + \frac{1}{3} \tan^3 x$ is the solution.

Ex. 11. Solve $xy - \frac{dy}{dx} = y^2 e^{-x^2}$.

[Bombay 58; Poona B.A. 60]

Solution. Write the equation as

$$y^{-2} \frac{dy}{dx} - xy^{-2} = -e^{-x^2}.$$

Put $-y^{-2} = v$ or $2y^{-3} \frac{dy}{dx} = \frac{dv}{dx}$.

$\therefore \frac{1}{2} \frac{dv}{dx} + xv = -e^{-x^2}$ or $\frac{dv}{dx} + 2xv = -2e^{-x^2}$.

Linear, I.F. = $e^{\int 2x dx} = e^{x^2}$,

$\therefore v \cdot e^{x^2} = C - 2 \int e^{-x^2} \cdot e^{x^2} dx = C - 2x$

or $-\frac{1}{y^2} e^{x^2} = C - 2x$ is the solution.

Ex. 12. Solve $3x(1-x^2)y^2 \frac{dy}{dx} + (2x^2-1)y^3 = ax^2$.

[Delhi Hons. 59]

Solution. The equation can be written as

$$3y^2 \frac{dy}{dx} + \frac{2x^2-1}{x(1-x^2)} y^3 = \frac{ax^2}{(1-x^2)}$$

or $\frac{dv}{dx} + \frac{2x^2-1}{x(1-x^2)} v = \frac{ax^2}{1-x^2}$ where $v = y^3$.

Linear, now $\int \frac{2x^2-1}{x(1-x^2)} dx = \int \frac{2x^2-1}{x(1-x)(1+x)} dx$
 $= \int \left(-\frac{1}{x} + \frac{\frac{1}{2}}{1-x} - \frac{\frac{1}{2}}{1+x} \right) dx$
 $= -\log x - \frac{1}{2} \log(1-x) - \frac{1}{2} \log(1+x)$
 $= -\log \{x(1-x^2)^{1/2}\}.$

Hence I.F. = $e^{\int P dx} = \frac{1}{x(1-x^2)^{1/2}}$ etc.

Ex. 13. Solve $x \frac{dy}{dx} - 2y = xy^4$.

[Gujrat 59]

Solution. Dividing by xy^4 , we get

$y^{-4} \frac{dy}{dx} - \frac{2}{x} y^{-3} = 1$. Put $-y^{-3} = v$, $3y^{-4} \frac{dy}{dx} = \frac{dv}{dx}$.

$\therefore \frac{dv}{dx} + \frac{6}{x} v = 3$. Linear, I.F. = $e^{\int \frac{6}{x} dx} = x^6$.

Hence $vx^6 = C + \int 3x^6 dx = C + \frac{3x^7}{7}$

or $-y^{-3}x^6 = C + \frac{3}{7}x^7$ is the solution.

Ex. 14. A curve passes through the point (0, 1) and the gradient at (x, y) on it is $y(xy-1)$. Find the equation. [Nagpur 61]

Solution. We have $\frac{dy}{dx} = y(xy-1)$

or $\frac{dy}{dx} + y = xy^2$ or $y^{-2} \frac{dy}{dx} + \frac{1}{y} = x$.

Put $-\frac{1}{y} = v, y^{-2} \frac{dy}{dx} = \frac{dv}{dx}$.

$\therefore$ equation is $\frac{dv}{dx} - v = x$, linear, I. F. = e^{-x} .

$\therefore ve^{-x} = \int xe^{-x} dx + C = -xe^{-x} - e^{-x} + C$

or $-\frac{1}{y} = -(x+1) + Ce^x$ or $\frac{1}{y} = (x+1) - Ce^x$.

It passes through (0, 1), i.e. when $x=0, y=1$. $\therefore 1 = 1 - C$ or $C=0$.

Therefore the curve is given by $\frac{1}{y} = (x+1)$ or $1 = y(x+1)$.

Ex. 15. Solve, $\sec^2 y \frac{dy}{dx} + (\tan y) 2x = x^3$

[Agra 63, 61; Gujrat 51; Delhi Hons. 64]

Solution. Putting $\tan y = v \therefore \sec^2 y \frac{dy}{dx} = \frac{dv}{dx}$, The equation becomes

$$\frac{dv}{dx} + 2xv = x^3$$

Linear. I. F. = $e^{\int 2x dx} = e^{x^2}$

$\therefore ve^{x^2} = C + \int x^3 e^{x^2} dx, x^2 = t, \therefore 2x dx = dt$

$\therefore x dx = \frac{1}{2} dt = C + \frac{1}{2} \int te^t dt = C + \frac{1}{2} e^t (t-1)$

$\therefore \tan y e^{x^2} = C + \frac{1}{2} e^{x^2} (\tan y - 1)$ [which is wrong]

The correct solution is $\tan y e^{x^2} = e + \frac{1}{2} e^{x^2} (x^2 - 1)$

Ex. 16. Solve $\frac{dy}{dx} + y \cos x = y^n \sin 2x$.

[Delhi Hons. 58]

Solution. Dividing by y^n , we get $y^{-n} \frac{dy}{dx} + y^{-n+1} \cos x = \sin 2x$.

Putting $y^{-n+1} = v, (1-n) y^{-n} \frac{dy}{dx} = \frac{dv}{dx}$, the equation becomes

$$\frac{dv}{dx} + v(1-n) \cos x = (1-n) \sin 2x.$$

I.F. = $e^{\int (1-n) \cos x dx} = e^{(1-n) \sin x}$.

$v \cdot e^{(1-n) \sin x} = C + \int e^{(1-n) \sin x} \cdot 2 \sin x \cos x dx$.

Now put $(1-n) \sin x = t$ and integrate.

Ex. 17. Solve the following linear equations :

(i) $(1+x^2) \frac{dy}{dx} + y = e^{\tan^{-1} x}$,

I. F. = $e^{\tan^{-1} x}$.

Ans. $y = \frac{1}{2} e^{\tan^{-1} x} + C e^{-\tan^{-1} x}$.

(ii) $x \cos x \frac{dy}{dx} + y(x \sin x + \cos x) = 1$

$$\therefore \frac{dy}{dx} + (\tan x + \frac{1}{x})y = \frac{1}{x \cos x} \quad [\text{Linear}]$$

$$P = \tan x + \frac{1}{x}$$

$$\text{I. F.} = e^{\int (\tan x + \frac{1}{x}) dx} = e^{-\log \cos x + \log x} = e^{\log x \sec x} \\ = \log x \sec x \text{ which is wrong solution.}$$

The correct solution is, I. F. = $x \sec x$.

$$(iii) \sin 2x \frac{dy}{dx} - y = \tan x.$$

$$P = -\operatorname{cosec} 2x. \text{ I.F.} = (\tan x)^{-1/2}$$

$$\text{Ans. } y = \tan x + C \sqrt{(\tan x)}.$$

[Agra 59]

$$(iv) x(x^2 + 1) \frac{dy}{dx} = x(1 - x^2) + x^3 \log x.$$

$$P = \frac{x^2 - 1}{x(x^2 + 1)}. \text{ I.F.} = \frac{x^2 + 1}{x}.$$

$$\text{Ans. } y = \frac{x^2 + 1}{x} + \frac{1}{2} x^2 \log x - \frac{1}{4} x^4 + C.$$

$$(v) \sqrt{a^2 + x^2} \frac{dy}{dx} + y = \sqrt{a^2 + x^2} - x$$

$$\therefore \frac{dy}{dx} + \frac{1}{\sqrt{a^2 + x^2}} y = 1 - \frac{x}{\sqrt{a^2 + x^2}} \text{ which is Linear form.}$$

$$\text{I. F.} = \frac{1}{\sqrt{a^2 + x^2}}, \quad \text{I. F.} = e^{\int \frac{1}{\sqrt{a^2 + x^2}} dx}$$

$$= e^{\log [x + \sqrt{x^2 + a^2}]} = \frac{x + \sqrt{a^2 + x^2}}{a} \text{ which is wrong}$$

The correct solun. is I. F. = $x + \sqrt{a^2 + x^2}$

$$(vi) x \log x \frac{dy}{dx} + y = 2 \log x$$

[Punjab 56, 54]

$$\text{I. F.} = \log x.$$

$$\text{Ans. } y \log x = (\log x)^2 + C.$$

[Raj. B.sc. 59]

$$(vii) \frac{dy}{dx} + \frac{y}{(1-x)\sqrt{x}} = 1 - \sqrt{x}.$$

$$\text{I. F.} = \frac{1 + \sqrt{x}}{1 - \sqrt{x}}$$

$$\text{Ans. } y \cdot \frac{1 + \sqrt{x}}{1 - \sqrt{x}} = x + \frac{2}{3} x^{3/2} + C.$$

[Raj. B. Sc. 56]

$$(viii) \frac{dy}{dx} + y \tan x = \sec x.$$

$$\text{I. F.} = e^{\log \sec x} = \sec x.$$

$$\text{Ans. } y \sec x = \tan x + C.$$

$$(ix) \cos^2 x \frac{dy}{dx} + y \tan x = \sin x \cos x - \cos^2 x.$$

[Punjab 56]

$$\text{I. F.} = e^{\tan x}.$$

$$(x) x(1 - x^2) dy + (2x^2 y - y - ax^3) dx = 0. \quad \text{Ans. } y e^{\tan x} = e^{\tan x} (\tan x - 1) + C.$$

[Nag. 62; Agra 60; Cal. Hons. 61]

$$\text{Equation is } \frac{dy}{dx} + y \frac{2x^2 - 1}{x(1 - x^2)} = \frac{ax^2}{1 - x^2}. \quad \text{I. F.} = \frac{1}{x\sqrt{1 - x^2}}$$

$$(xi) (1 - x^2) dy + y dx = a dx.$$

[Cal. Hons. 62]

$$\text{Equation is } \frac{dy}{dx} + \frac{y}{1 - x^2} = \frac{a}{1 - x^2}. \quad \text{I. F.} = \frac{1}{\sqrt{1 - x^2}}.$$